KESTREL scrap build guide: S1-Scrap and K1-Scrap

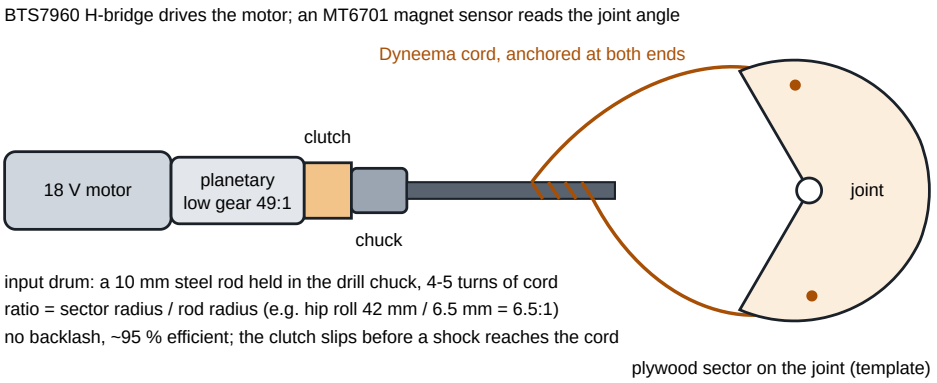
This guide builds a full-size, 1.75 m humanoid from salvaged parts and hand tools, with no CNC and no 3D printer. Every actuator is a cordless-drill motor and its gearbox, driving the joint through a capstan you make from a steel rod, Dyneema cord and a plywood sector. The robot is sized by the same simulation as K1 and S1.

There are two versions. Build S1-Scrap first; K1-Scrap adds to it.

	S1-Scrap	K1-Scrap
Target cost	A$1,000	A$1,000
Parts list	**A$917 buying the cheapest**, A$492 scrounging (../s1/generated/bom.md)	A$1,280 buying, **A$727 scrounging** (../k1/generated/bom.md)
Height / mass	1.75 m / about 30 kg	1.75 m / about 39 kg
Actuated joints	16: legs 2 × 6, shoulder pitch and elbow per arm	25: legs 2 × 6, waist yaw, arms 2 × 4, neck 2; plus a one-servo gripper per hand
Motors	18 drill gearmotors (two per knee)	23 drill gearmotors (two per knee), 4 hobby servos
Walking	**0.25 m/s, always balanced** (0.36 m/s fast)	**0.25 m/s** (0.36 m/s fast)
Steps, objects	15 cm steps, 3 kg box, 0.5 kg per hand	15 cm steps, 5 kg box, 1 kg per hand
Looks	bare frame, webcam head	old-phone face, EVA-foam armour in K1 white / graphite / amber

S1-Scrap fits A$1,000 even if you buy every part. If you scrounge, it costs about half that. **On A$500:** set `SCRAP = SLOW` in `calc/kestrel/variant_s1scrap.py` and re-run step 11's scripts. That version has one drill per knee (16 drills), walks at 0.12 m/s with 10 cm steps, and its parts come to about A$463 scrounged.

What these robots can and cannot do. A drill gearmotor is strong for its price, but it reaches its torque through a large gear ratio. The motor's spinning rotor, multiplied by the ratio squared, feels like 2–4 kg·m² of inertia at the knee, about 20 times the leg's own. So the scrap robots cannot walk the fast, bouncy way commercial humanoids do. They walk the slow, always-balanced way early research robots did: shift the weight over one foot, step, shift again. Everything in this guide is sized for that. The simulation that sizes them, and the torque, speed and missed-step checks, are in ../s1/generated/actuator_selection.md and ../k1/generated/actuator_selection.md.



Build time	about 150–250 hours for S1-Scrap, 250–350 for K1-Scrap, plus the time to collect parts (or 2–4 weeks of delivery if you buy)
Skills	hacksaw, drill press, tapping, soldering, crimping, flashing an ESP32
Tools (you have them)	drill press with a 22 mm and a 25 mm Forstner bit or hole saw, hacksaw, jigsaw, files, tap set M3–M8, hex keys, a spring or luggage scale, multimeter, soldering iron, bench supply with current limit

Step 0: Safety

1. **Salvaged lithium cells are the main hazard.** Test every cell (step 7). Never use a swollen, dented, hot or low-voltage cell. Charge only supervised, on a concrete floor or in a steel box, never overnight.
2. **The joints are strong.** The knee can push 134–183 N·m. Keep fingers out of capstans and joints whenever the motors are powered.
3. **E-stop first.** The hard-wired e-stop (step 8) works before any motor moves on the robot.
4. **Gantry always.** Hang the robot from a garage beam or an A-frame by chest and pelvis straps with the rope just slack, for every powered test.
5. **Two people** from the first powered motion: one holds the e-stop and does nothing else.

Step 1: Collect the parts

The parts lists give two prices per line. "Scrounge" assumes you collect the salvageable parts for little or nothing:

Part	Where to scrounge it
18 V cordless drills with dead batteries (18 or 23)	council kerbside collections, tip shops, Marketplace "broken drill", friends and family. Ask builders: dead-battery drills are thrown out all the time. The battery is not needed
18650 cells (25–30 good ones)	dead laptop packs, power-tool packs, e-bike packs (e-waste days, computer repairers). Expect 30–50 % of cells to pass the test
Aluminium angle and tube	metal recyclers, sold by weight at a fraction of the hardware-store price
12 mm plywood	offcuts from builders, cabinet makers, the hardware-store offcut bin
Cables, connectors	old PC power supplies and appliances

Buy new, from AliExpress (allow 2–4 weeks): BTS7960 H-bridges, MT6701 angle sensors, TCA9548A multiplexers, ESP32-S3 boards, a BNO085 IMU, 608 bearings, 3 mm Dyneema cord, a 5S BMS, the charger buck module. K1-Scrap adds four 20 kg·cm servos and a PCA9685 board.

Do not mix drill models: all 18 (or 23) should be the same model if you can. Then one clutch calibration, one clamp design and one firmware setting cover all of them.

Check: every line of the parts list is in a box, and every drill spins when fed 18 V from the bench supply.

Step 2: Test and convert the drills

The simulation assumes a 550-size 18 V motor and a two-stage planetary of about 49:1 in low gear (400 rpm). Measure yours before you build anything, because the numbers decide the capstan sizes.

1. **Strip each drill.** Remove the housing screws. Keep the motor, the gearbox, the clutch ring, the chuck and the speed-selector slide. Discard the trigger switch and the battery terminals.

2. **Lock low gear.** Slide the selector to low (the slow, strong setting) and fix it with a screw or epoxy, so it can never jump to high gear.

3. **Make a clamp.** Two 12 mm plywood plates with a hole bored to the gearbox diameter, clamped round the gearbox with 4 × M5 bolts, and a strap over the motor. The clamp bolts to the frame.

4. **Lever test** (the same as S1's): bench supply at 18 V with the current limit at 40 A or the highest it has, a 0.5 m aluminium bar in the chuck, the bar resting on a luggage scale.

   - Record the force at 10, 20, 30 and 40 A. Torque = force (kg) × 9.81 × 0.5 m. At 40 A the model expects about 13 N·m at the chuck.
   - Record the no-load speed at 18 V: about 400 rpm (42 rad/s).
   - Note your numbers in `scrap/data/actuators.json` (`Kt`, `R`, `gear`) and re-run `python scrap/calc/scrap_select.py --variant s1scrap`. It re-sizes every capstan from your measurements.

5. **Calibrate the clutch.** Each drive has a clutch setting in the table below (N·m at the chuck). With the lever and the scale, find the clutch-ring number that slips at that torque, and fix the ring there with a drop of epoxy. The clutch is the mechanical fuse: it slips before the shock of a missed step can snap a cord.

Check: every drill gives within 15 % of the model torque and speed, and every clutch slips at its set torque.

Step 3: Make the capstan drives

A capstan is a cord wrapped 4–5 times around a small drum (the steel rod in the drill chuck), with both ends anchored on a large sector bolted to the joint. Turning the rod winds the cord on one side and off the other, and turns the sector. The ratio is the sector radius divided by the rod radius. There is no backlash and nothing to machine.

S1-Scrap drives (from ../s1/generated/actuator_selection.md):

Joint	Motors	Ratio	Sector template	Joint peak / no-load speed	Clutch at the chuck
Hip yaw	1	2.0:1	`s1/templates/sector_X-HY.svg`	24 N·m / 21 rad/s	21 N·m
Hip roll	1	6.5:1	`s1/templates/sector_X-HR.svg`	80 N·m / 6.5 rad/s	16 N·m
Hip pitch	1	3.5:1	`s1/templates/sector_X-HP.svg`	43 N·m / 12 rad/s	23 N·m
Knee	**2**	5.5:1, **2 cords**	`s1/templates/sector_X-KN.svg`	134 N·m / 7.6 rad/s	23 N·m each
Ankle (2 per foot)	1 each	3.25:1	`s1/templates/sector_X-AN.svg`	40 N·m at the crank	23 N·m
Shoulder pitch, elbow	1	direct	— (chuck holds the joint shaft)	13 N·m	22 / 15 N·m

K1-Scrap drives (from ../k1/generated/actuator_selection.md):

Joint	Motors	Ratio	Sector template	Joint peak / no-load speed	Clutch at the chuck
Hip yaw	1	2.75:1	`k1/templates/sector_K-HY.svg`	34 N·m / 15 rad/s	23 N·m
Hip roll	1	8.25:1	`k1/templates/sector_K-HR.svg`	101 N·m / 5.1 rad/s	16 N·m
Hip pitch	1	4.5:1	`k1/templates/sector_K-HP.svg`	55 N·m / 9.3 rad/s	23 N·m
Knee	**2**	7.5:1, **2 cords**	`k1/templates/sector_K-KN.svg`	183 N·m / 5.6 rad/s	22.6 N·m each
Ankle (2 per foot)	1 each	4.25:1	`k1/templates/sector_K-AN.svg`	52 N·m at the crank	23 N·m
Waist yaw	1	1.5:1	`k1/templates/sector_K-WY.svg`	18 N·m	21 N·m
Shoulder pitch	1	1.75:1	`k1/templates/sector_K-SP.svg`	21 N·m	21 N·m
Shoulder roll, yaw	1	direct	—	13 N·m	6 / 10 N·m
Elbow	1	1.5:1	`k1/templates/sector_K-EL.svg`	18 N·m	17.5 N·m

Joint	Motors	Ratio	Sector template	Joint peak / no-load speed	Clutch at the chuck
Neck yaw, pitch	servo	direct	—	1.9 N·m (stall)	—

1. **Print the templates at 100 %** ("actual size", no fit-to-page). Check the 50 mm bar with a ruler.
2. **Cut the sectors.** Glue each template to 12 mm plywood with a glue stick. For the knee, glue two layers of plywood together first (24 mm), or use 6 mm aluminium plate. Drill the holes on the drill press, cut the outer contour with a jigsaw, and file or sand to the line.
3. **Cut the cord groove.** Clamp the sector on an M8 bolt in the drill press and hold a small round file against the edge while it turns, to make a 3 mm deep groove on the dashed line. Both knees have two grooves 6 mm apart.
4. **Input drums.** Cut a 60 mm length of 10 mm steel rod per drive and chuck it, 40 mm out of the chuck. Where the template says "25 mm tube", sleeve a 25 mm aluminium or steel tube over the rod with epoxy and a cross pin. A bigger drum is kinder to the cord.
5. **Bearings.** Every joint turns on 608 bearings: bore a 22 mm seat with the Forstner bit 7 mm deep in plywood (a light press fit) and run an M8 bolt through as the axle. Support the free end of each input rod with a 608 in a plywood plate, so the cord tension does not bend the chuck.
6. **Wind the cord.**
   - Tie one end through the first anchor hole.
   - Lay the cord along the groove to the rod, wrap it 4–5 turns round the rod, and take it back along the groove to the second anchor.
   - Tension it with the M4 anchor screw until the cord is tight enough to "ping" when plucked. Re-tension after the first hour of running.
   - For two cords, wind two separate loops, one in each groove, and tension them equally.
7. **Ankles.** Each ankle motor drives a 40 mm crank through its capstan. A push-rod (M8 threaded rod with rod ends) runs from the crank to a lever 52 mm behind the ankle axis, as on S1. Two motors per foot give pitch (both rods together) and roll (rods opposite).

Check: every joint moves through its range by hand (motor unpowered, clutch slipping is fine) with no rub and no cord slip on the rod.

Step 4: Build the frame

The frame follows S1's skeleton, so the simulation applies. Key dimensions (centre to centre, from `calc/kestrel/params.py`):

	mm
Hip joints apart	210 (105 each side of the centre)
Hip-yaw axis above the hip centre	75
Thigh (hip to knee), shank (knee to ankle)	420, 420
Ankle axis above the floor	75
Foot	260 long × 110 wide, ankle 70 mm from the heel
Waist above the hip centre (fixed on S1-Scrap)	100 + 70
Shoulders above the waist, apart	335, 400 (200 each side)
Upper arm, forearm	300, 260
Neck above the waist	400

1. **Cut the members** from 25 × 25 × 3 mm aluminium angle (pelvis, torso ladder, brackets) and 40 × 40 mm square tube (thighs and shanks) with the hacksaw. Square the ends with a file against a square.

2. **Joint plates** are 12 mm plywood: mark them out from the dimensions, drill on the drill press, cut with the jigsaw. Bolt them to the aluminium with M5 bolts and nyloc nuts. Plywood is light, takes a 608 press fit, and is easy to remake if a part is wrong.

3. **Hip.** The hip-yaw drive sits upright in the pelvis; the hip-roll drive hangs below it; the hip-pitch drive sits beside the hip-roll bracket, its capstan sector on the thigh. Lay out each joint so its sector clears its neighbours over the full range in the table at the end of this step.

4. **Knee.** The two knee drills bolt side by side to the top of the thigh tube. Their rods are joined on one common shaft with a slotted tube, so the motors share the load, and the shaft runs down the side of the thigh to the double-groove sector bolted to the shank at the knee axis.

5. **Feet.** 12 mm plywood stiffened with aluminium angle along both edges, a rubber sole from an old tyre tube or door mat, and four microswitches (heel and toe, each side) under the sole as contact sensors.

6. **Torso.** An aluminium-angle ladder with plywood back and shoulder plates. The battery and electronics bolt to the back plate; keep them low and central.

Joint ranges (degrees): hip yaw −25…+35, hip roll −20…+25, hip pitch −100…+30, knee 0…120, ankle pitch −40…+40, ankle roll −20…+20, shoulder pitch −180…+60, elbow −130…0. K1-Scrap: shoulder roll −10…+150, shoulder yaw ±90, waist yaw ±60, neck yaw ±80, neck pitch −30…+40. Fit a rubber stop at each end of every range.

Check: weigh every sub-assembly as you finish it and write it down. S1-Scrap's model expects about 30 kg in all, K1-Scrap's about 39 kg. If you are heavier, re-run the selection with the measured masses (step 11).

Step 5: Wire every actuator

1. Each drill motor goes to one **BTS7960**: motor to M+ and M−, the 18 V bus to B+ and B−, and to the ESP32 RPWM, LPWM, R_EN and L_EN.
    - Fit a 1 kΩ resistor from R_IS and L_IS to ground. The IS voltage is the motor current / 8.5 A (R11): the firmware reads it to limit current to 40 A peak and 14 A continuous.
2. Each joint gets an **MT6701** angle sensor facing a diametric magnet glued to the end of the joint axle. Sensors go through **TCA9548A** multiplexers (8 per board), because every MT6701 has the same I²C address.
3. **Controllers:** S1-Scrap uses two ESP32-S3 boards (left leg and arm, right leg and arm); K1-Scrap adds a third for the waist, neck and grippers. They talk to each other over one UART bus.
4. Label both ends of every cable.

Step 6: Build the battery

1. **Test every salvaged cell.**
   - Reject any below 2.5 V, or that is swollen, dented, rusty or gets warm.
   - Charge each to 4.2 V, wait a week, and reject any that dropped below 4.1 V.
   - Measure capacity with a cheap charger-tester if you have one, and keep cells within 10 % of each other.
2. **Assemble a 5S pack:** 5 groups in series, 5 cells per group in parallel for S1-Scrap (5S5P, 18 V, about 12 Ah) and 6 per group for K1-Scrap (5S6P). Use plastic 18650 holders and 12 AWG wire, or a nickel-strip kit. Put the 5S BMS on the balance taps.
3. **Charger:** a CC/CV buck module set to 21.0 V and 2 A, fed from an old 24 V laptop or printer supply. Charge supervised only.
4. **Pack box:** plywood with a steel or aluminium liner, low in the torso.

Step 7: Power and the e-stop

1. **Power path:** battery → 40 A main fuse → **40 A automotive relay** (the contactor) → 18 V bus → a 15 A fuse per limb.
2. **E-stop loop:** the relay coil is fed through the latching mushroom button and a watchdog transistor the ESP32 holds on. Press the button, or let the firmware stop, and every motor loses power at once.
3. **Logic power:** a 5 V buck straight from the battery side of the relay, so an e-stop does not reset the controllers.
4. **First power-up with the bench supply** at 18 V and 2 A, no motors connected: check the e-stop opens the relay and the controllers stay on.

Check: the e-stop stops everything from every state; insulation from the bus to the frame is over 1 MΩ.

Step 8: Firmware

Each ESP32 runs, at 500 Hz per joint:

1. Read the joint angle (MT6701) and the motor current (BTS7960 IS pin).
2. **Position control:** duty = kp × (target − angle) − kd × speed, limited so the current never exceeds 40 A peak, and an I^2t estimate never exceeds 14 A continuous. Start with kp low and raise it until the joint holds without oscillating.
3. **Soft limits** 3° inside the rubber stops.
4. **Watchdog:** if a controller stops hearing from the others for 50 ms, it drops the e-stop transistor.
5. **Calibration:** hang the robot in the zero pose (legs straight, feet flat, arms down) and record each joint's sensor offset.

Step 9: Stand, then walk slowly, in the gantry

The simulation's slow walk is in `scrap/s1/data/gait_W1.csv` (K1-Scrap: `scrap/k1/data/gait_W1.csv`): joint angles in degrees, 100 rows per second, one column per joint. The fast walk is `gait_W2.csv`.

Stage	Setup	What to do	Pass
1	Hanging, feet off the ground	Move each joint to the CSV's first row, slowly	Holds the pose, no oscillation
2	Feet on the ground, rope taking most of the weight	Crouch to the first row and stand there	Holds for 5 minutes, motors warm, not hot
3	Rope slack	Shift the weight from foot to foot (hip roll and ankle roll only)	Stays balanced
4	Rope slack	Play `gait_W1.csv` at **quarter speed**	Steps without the rope taking load
5	Same	Half speed, then full speed	Repeatable, no falls into the rope
6	Rope still on	A 5 cm board, then 10 cm (S1-Scrap) or 15 cm (K1-Scrap)	Only after stage 5 is routine

Use the pelvis IMU to correct the ankle and hip roll angles a little each step (keep the torso upright). Once the scripted walk works, the reinforcement-learning package in `train/` has policies for both scrap robots (`python play.py --variant s1scrap --run s1scrap`), trained against the same slow actuators.

Step 10: K1-Scrap extras

1. **Arms:** four drives per arm (shoulder pitch, roll and yaw, then the elbow), from the K1-Scrap table.
2. **Waist yaw:** the torso sits on a 150 mm lazy-Susan turntable bearing, turned by a drill through its capstan.
3. **Grippers:** three plywood fingers on M3 pivot bolts, closed by fishing-line tendons wound on a servo horn and opened by springs, with rubber finger pads.
4. **Head:** an old smartphone in an EVA-foam helmet on the two neck servos. The phone is the camera, a second IMU and the face: show eyes on its screen. It connects to the ESP32s over Wi-Fi or USB.
5. **Armour:** cut EVA-foam floor tiles to paper patterns, heat them with a heat gun to curve them, glue the seams with contact cement, seal with primer, and paint K1 white with graphite joints and amber light stripes. Strap the armour on with Velcro so it comes off for maintenance.

Step 11: Change it

- **Measured masses or motors:** put them in `calc/kestrel/variant_s1scrap.py` (or `variant_k1scrap.py`) and `scrap/data/actuators.json`, then run `python scrap/calc/scrap_select.py --variant s1scrap`, `python scrap/calc/scrap_bom.py` and `python scrap/calc/scrap_templates.py`. You get new capstan sizes, clutch settings, templates and a new parts list.
- **Add a component:** add a line to `scrap/s1/data/add_ons.json` (same format as S1's) and re-run the selection.
- **Free hoverboards?** Each one gives two hub motors, an FOC motor board, a 36 V pack and a charger. The selection already knows the hub motor (`hub36`, on a bike-chain reduction). At the hoverboard board's stock current, drills win at every joint; with the board's current limit raised and measured, add `("hub36", 1)` to `ALLOWED` and try.
- **More speed:** a 6S pack (21.6 V) gives 20 % more speed if your motors tolerate it. Set `bus_V` and re-run.

Limits

- **Slow walking only.** The drives' reflected inertia rules out dynamic walking and running. A missed step makes the clutches slip and the robot sit down into the gantry harness; that is the design, not a fault.
- **Everything rests on step 2's measurements.** Re-run the selection with your numbers before cutting the sectors.
- **The salvaged parts are estimates.** Prices and motor constants are marked V, Vs or E in `docs/research/R11-scrap-build-parts.md`.
- **Safety as for K1 and S1:** gantry, e-stop in hand, nobody within reach until the robot has proven itself.